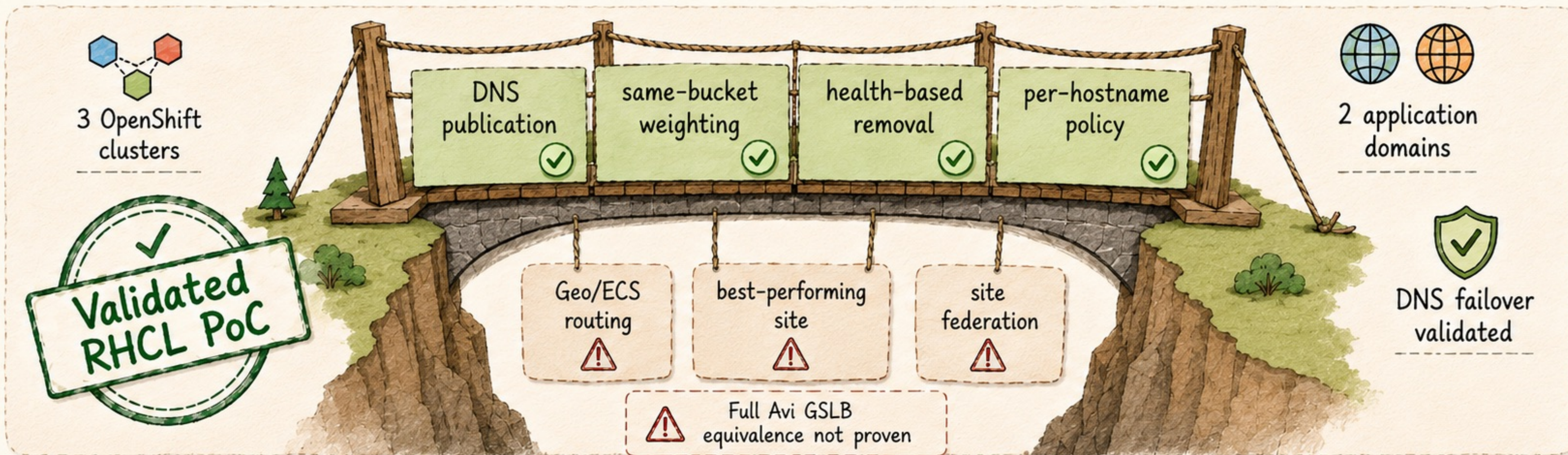


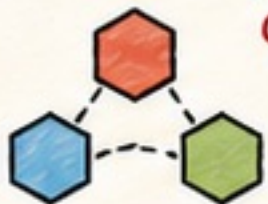
RHCL Three-Cluster GLB E2E Validation

Can RHCL replace Avi-style multi-geo, multi-cluster load balancing?



1

3 OpenShift
clusters



Validated RHCL PoC
spans three sites.

2

2 application
domains



Independent hostnames
can have separate policy.

3

DNS failover
validated



Failed clusters can be
removed from DNS answers.

Assessment Summary



Validated:

DNS publication, weighting,
health-based removal, per-hostname policy

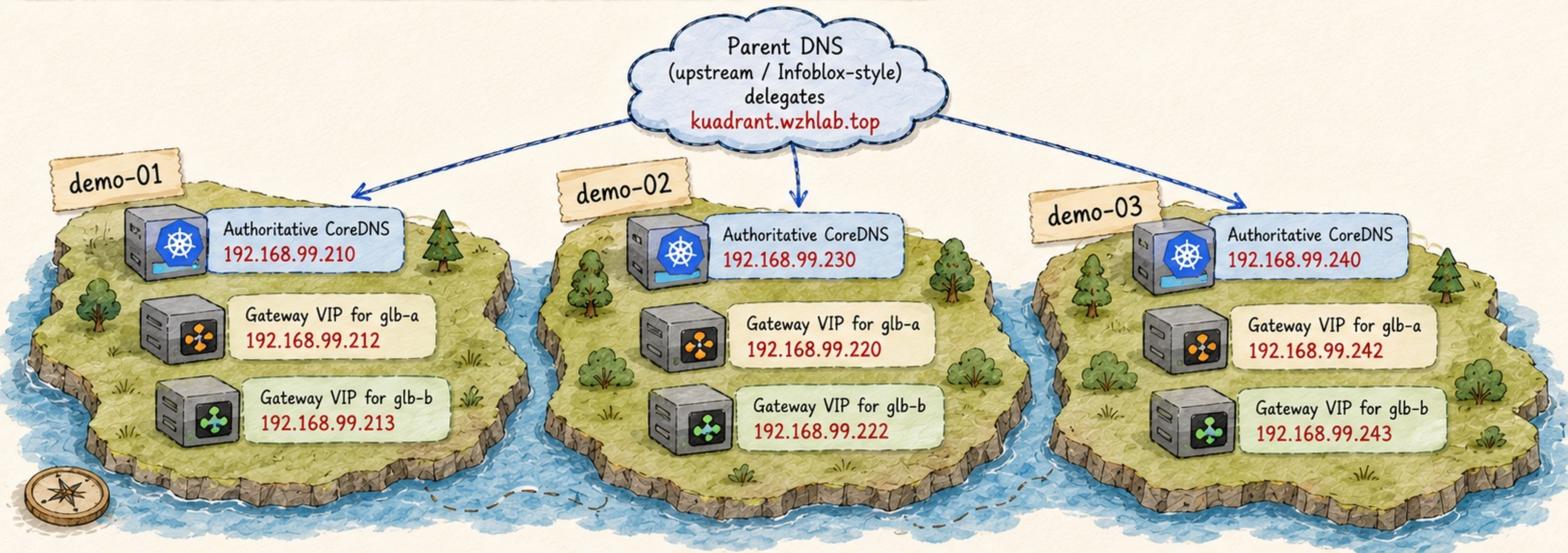


Not proven:

full Avi-style geo, ECS, latency,
and federation equivalence

Test Topology: 3 Clusters, 2 Domains

Parent DNS delegates `kuadrant.wzhlab.top` to RHCL-hosted CoreDNS VIPs.



Authoritative CoreDNS
serving RHCL records



Gateway VIP
for glb-a hostname



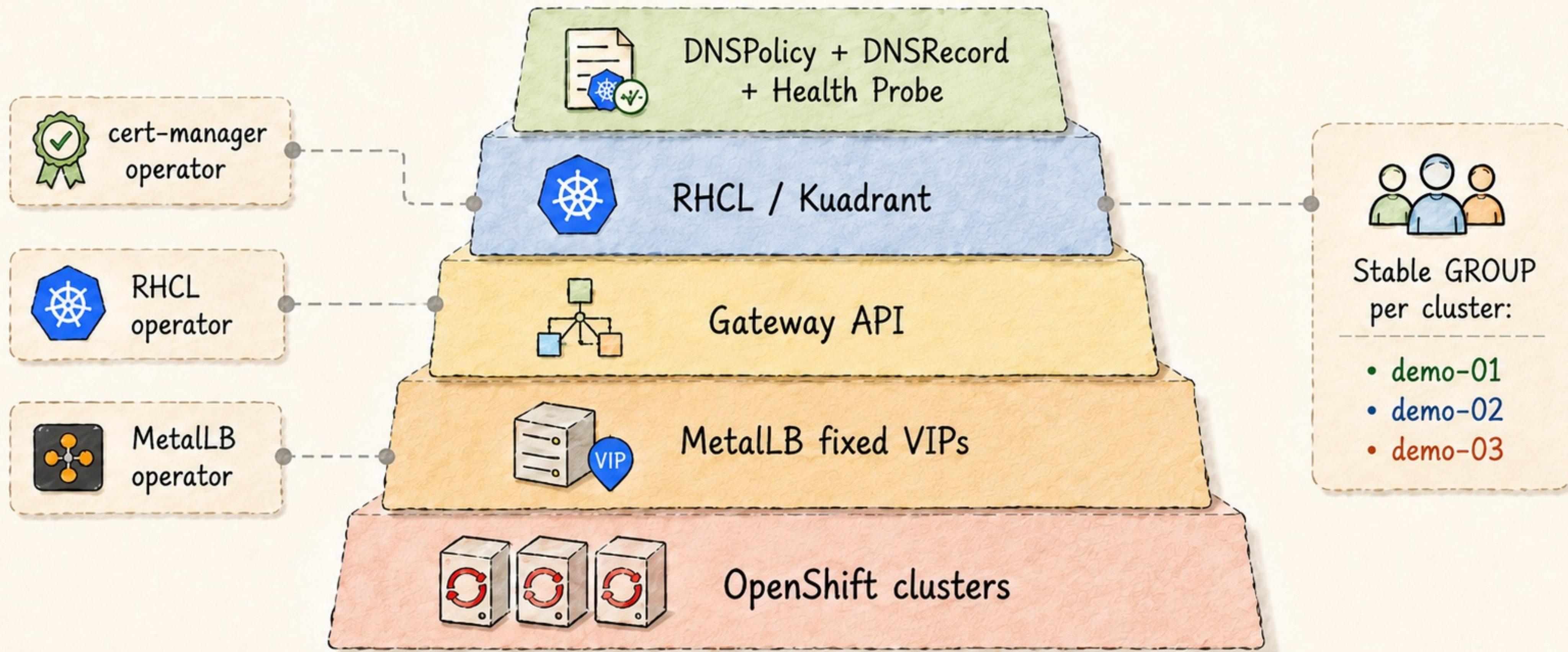
Gateway VIP
for glb-b hostname



Independent
hostname policy

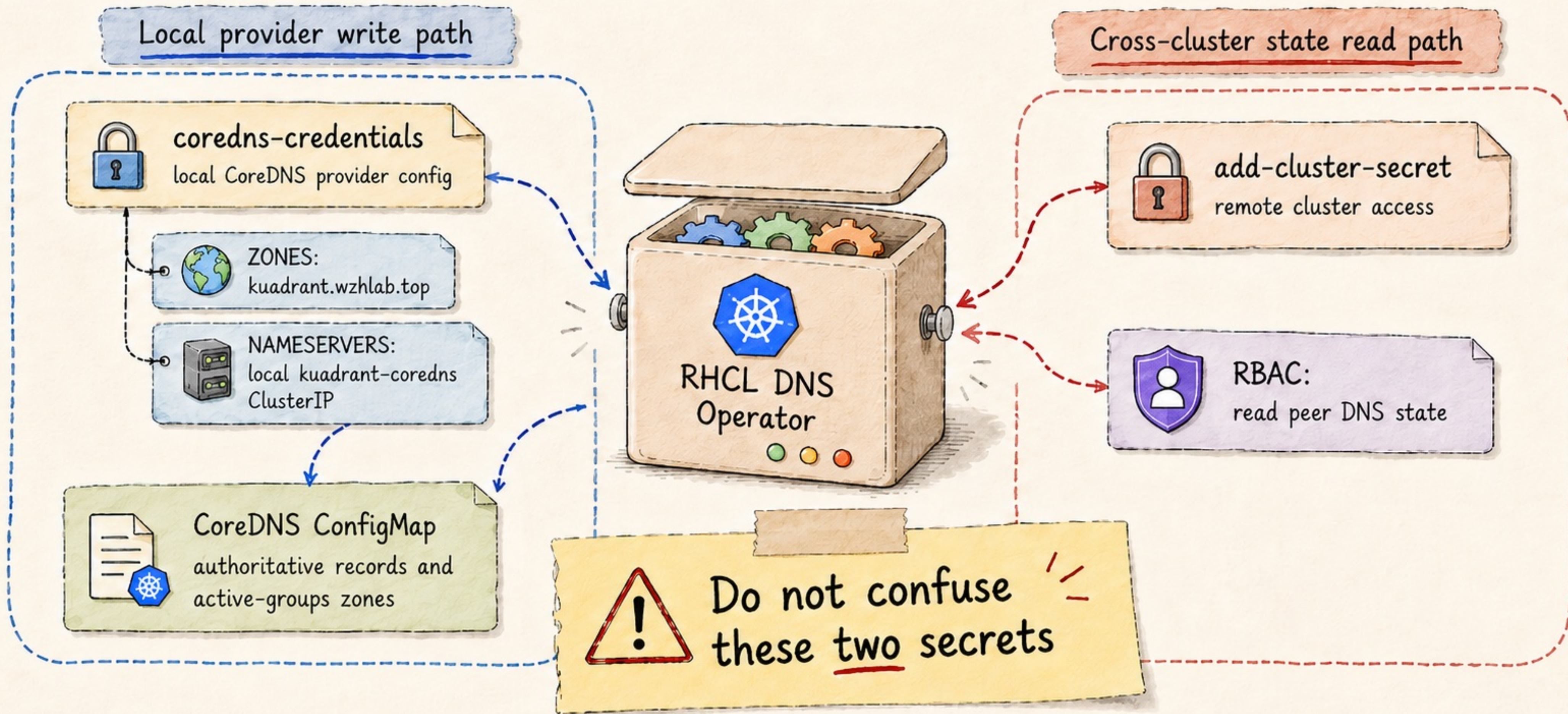
OpenShift-Native Foundation

Operators and Gateway API create the RHCL control plane.



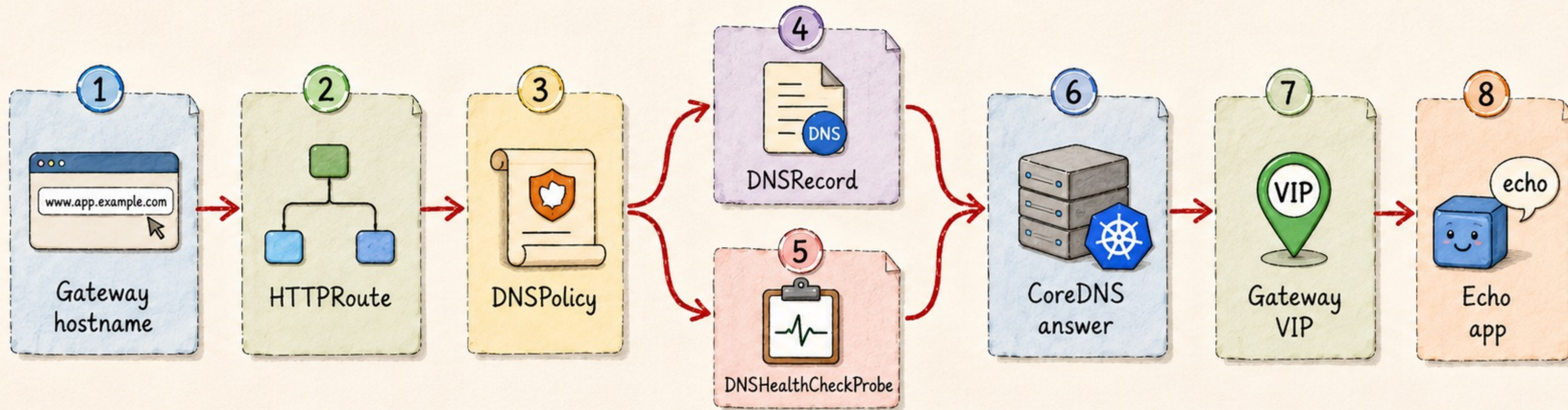
DNS Control Plane Breakdown

Provider configuration and peer-cluster access are different responsibilities.



How One Hostname Becomes a DNS Answer

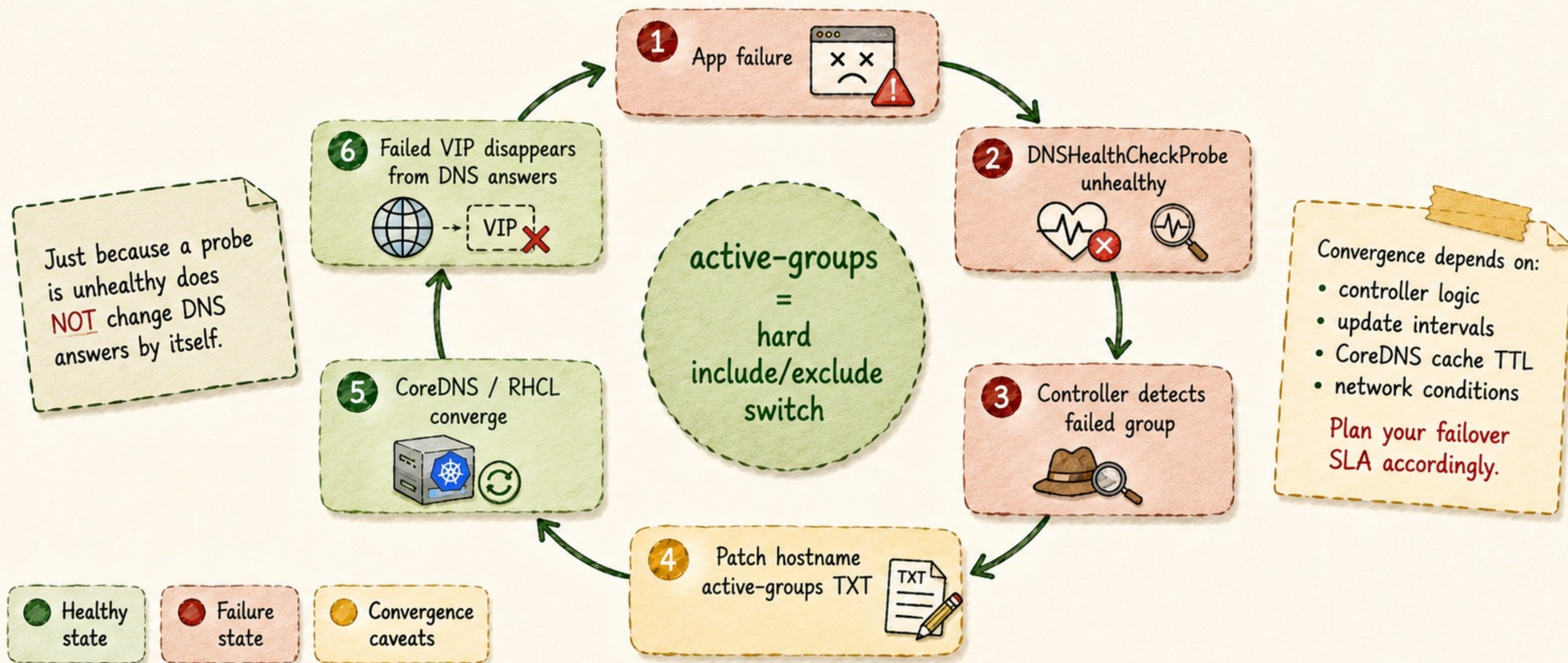
Gateway + HTTPRoute + DNSPolicy drives the RHCL DNS data path.



☆ $\text{Gateway hostname} + \text{DNSPolicy} = \text{DNSRecord} + \text{DNSHealthCheckProbe} + \text{CoreDNS answer}$

Failover Is a Membership Reconciliation Loop

Health signals must become DNS pool membership changes.



Evidence Model

The test watched desired state, health state, and real DNS answers.



1. Kubernetes objects

DNSPolicy, DNSRecord, DNSHealthCheckProbe, Deployment



DNSPolicy



DNSRecord



DNSHealth
CheckProbe



Deployment



2. DNS samples

repeated dig against
192.168.99.210, .230, .240

```
$ for ip in 192.168.99.210 192.168.99.230 192.168.99.240; do  
> dig +short app.kuadrant.wzhlab.top @$ip  
> done
```

@192.168.99.210

192.168.99.212
192.168.99.222
...

@192.168.99.230

192.168.99.220
192.168.99.242
...

@192.168.99.240

192.168.99.242
192.168.99.212
...



3. Failure action

scale one app down

```
$ oc scale deploy echo-app -n demo-02 --replicas=0
```



4. Convergence check

bad VIP removed from DNS answers

Before (unhealthy)

192.168.99.220 ✗



After (converged)

192.168.99.242 ✓

192.168.99.212 ✓



Bad VIP
gone from
DNS
answers



5. Restore action

app replicas and active-groups
returned to healthy state

```
$ oc scale deploy echo-app -n demo-02 --replicas=3
```



Active-groups TXT

demo-01	✓
demo-02	✓
demo-03	✓

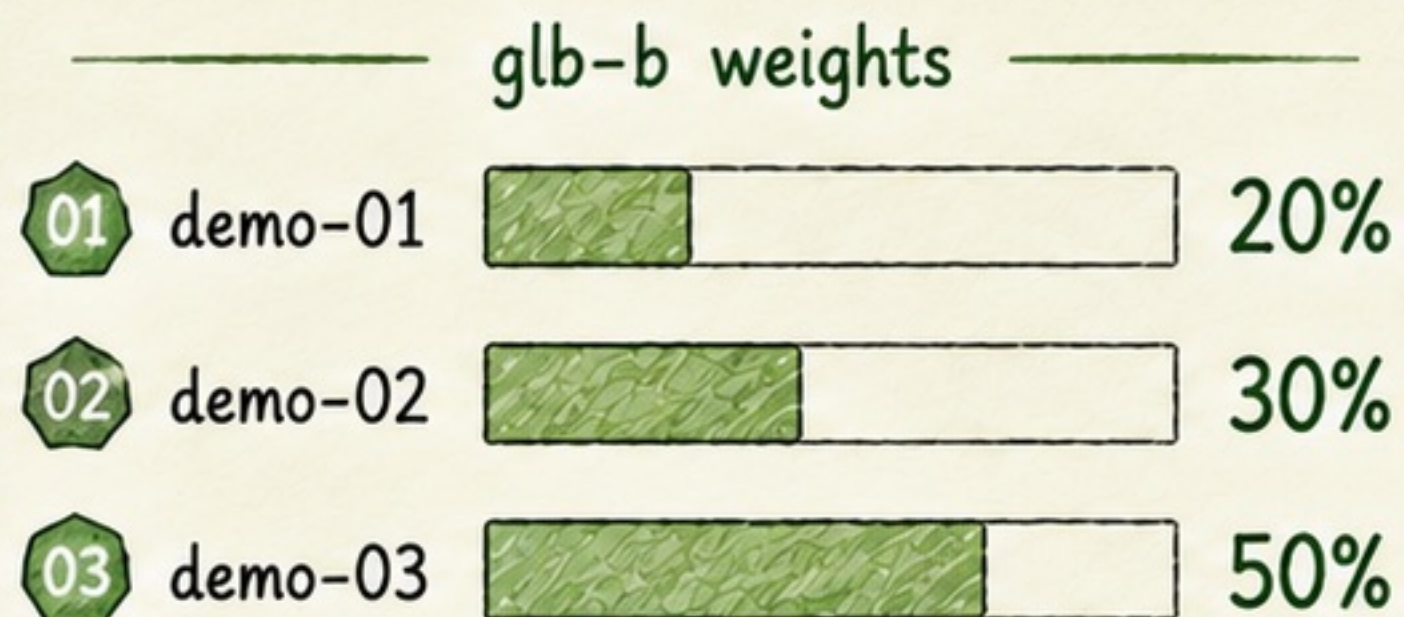
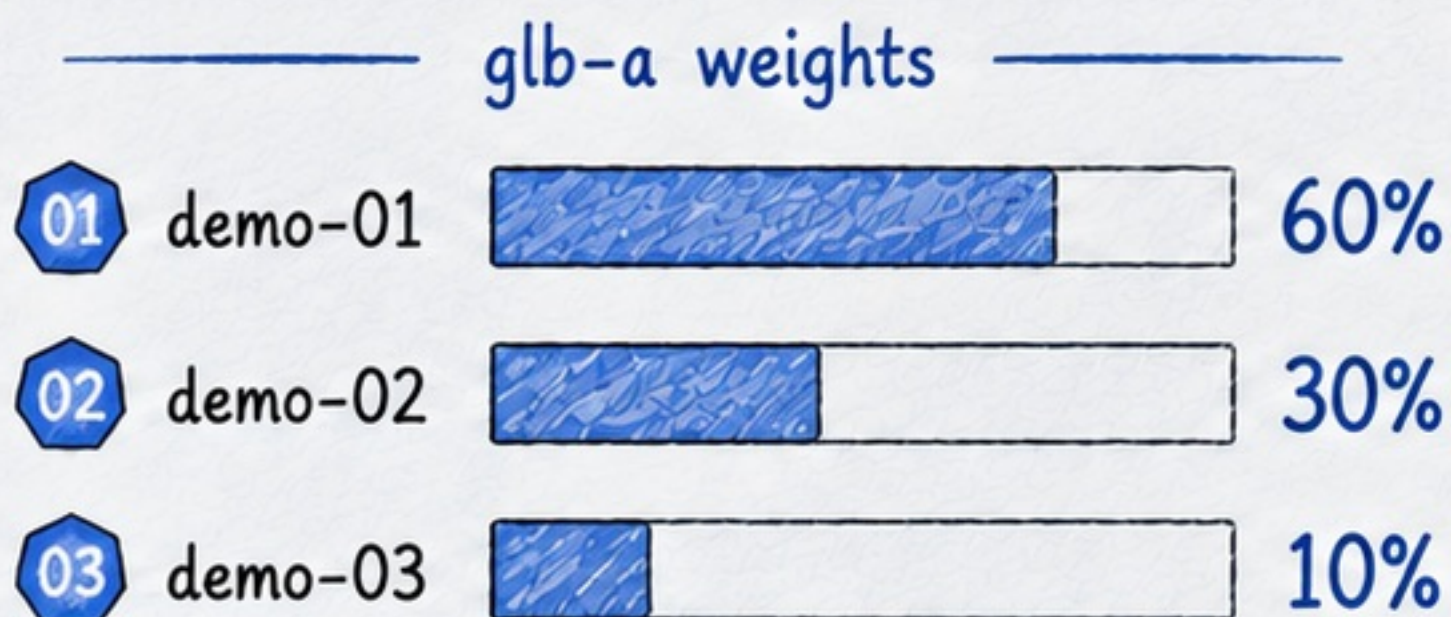
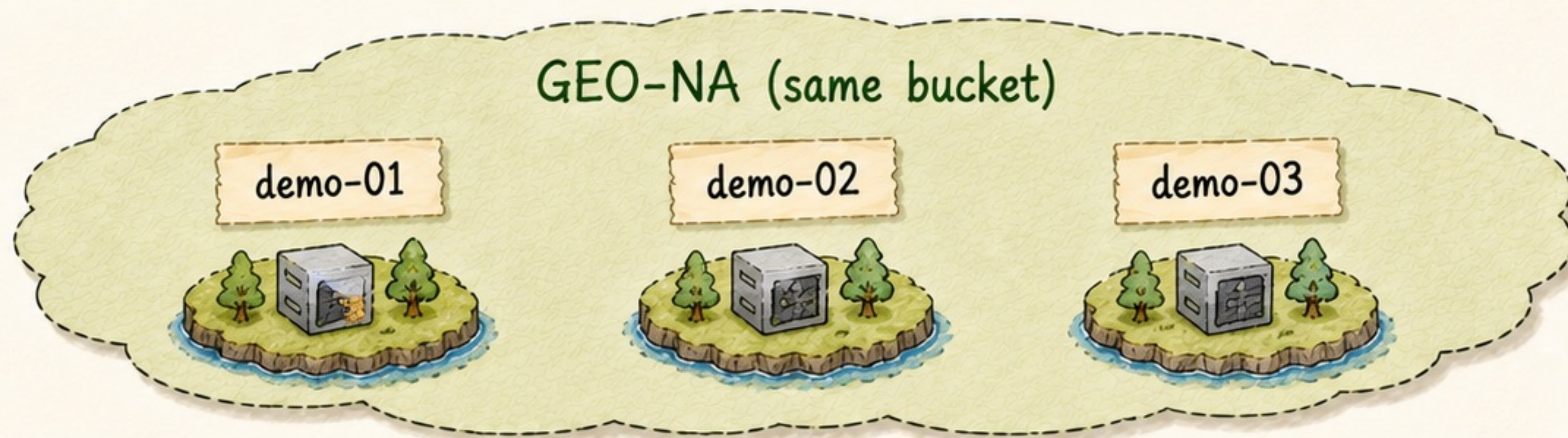


We validated control-plane intent, health signals, and real DNS data-plane behavior.



Scenario 1: Same-Bucket Weighting

All three clusters were in GEO-NA, with different weights per hostname.



Result:
same-bucket
weighted DNS
behavior validated



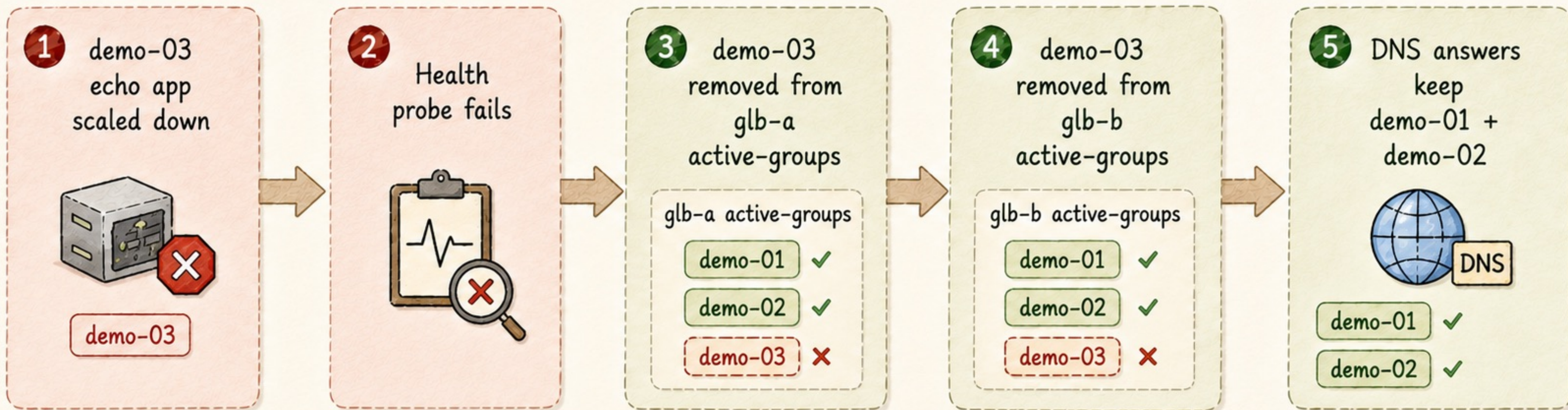
Interpretation:
weight is useful
inside a bucket



This scenario proves that weights influence distribution within the same GEO bucket.

Scenario 1 Failure: demo-03 Removed

Both hostnames removed the failed cluster from DNS answers.



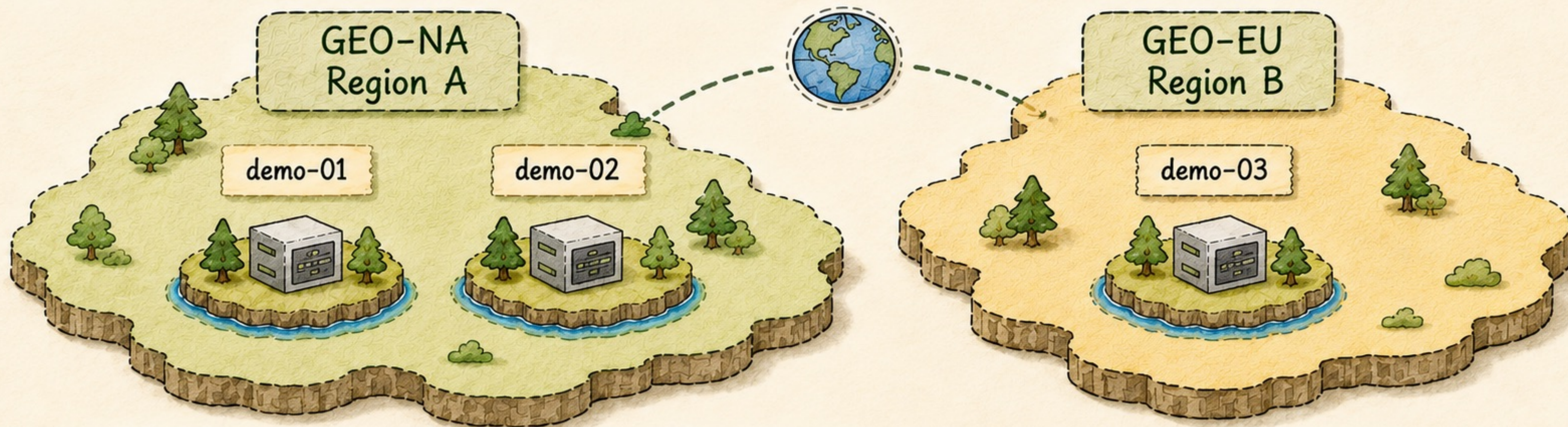
Validated:

failed Gateway VIP disappeared from both hostname pools

glb-a:	demo-01 ✓	demo-02 ✓
glb-b:	demo-01 ✓	demo-02 ✓

Scenario 2: Two Regions, Three Clusters

Two clusters in Region A; one cluster in Region B.



glb-a policy

	Region A	70/20
	Region B	10

glb-b policy

	Region A	20/50
	Region B	30

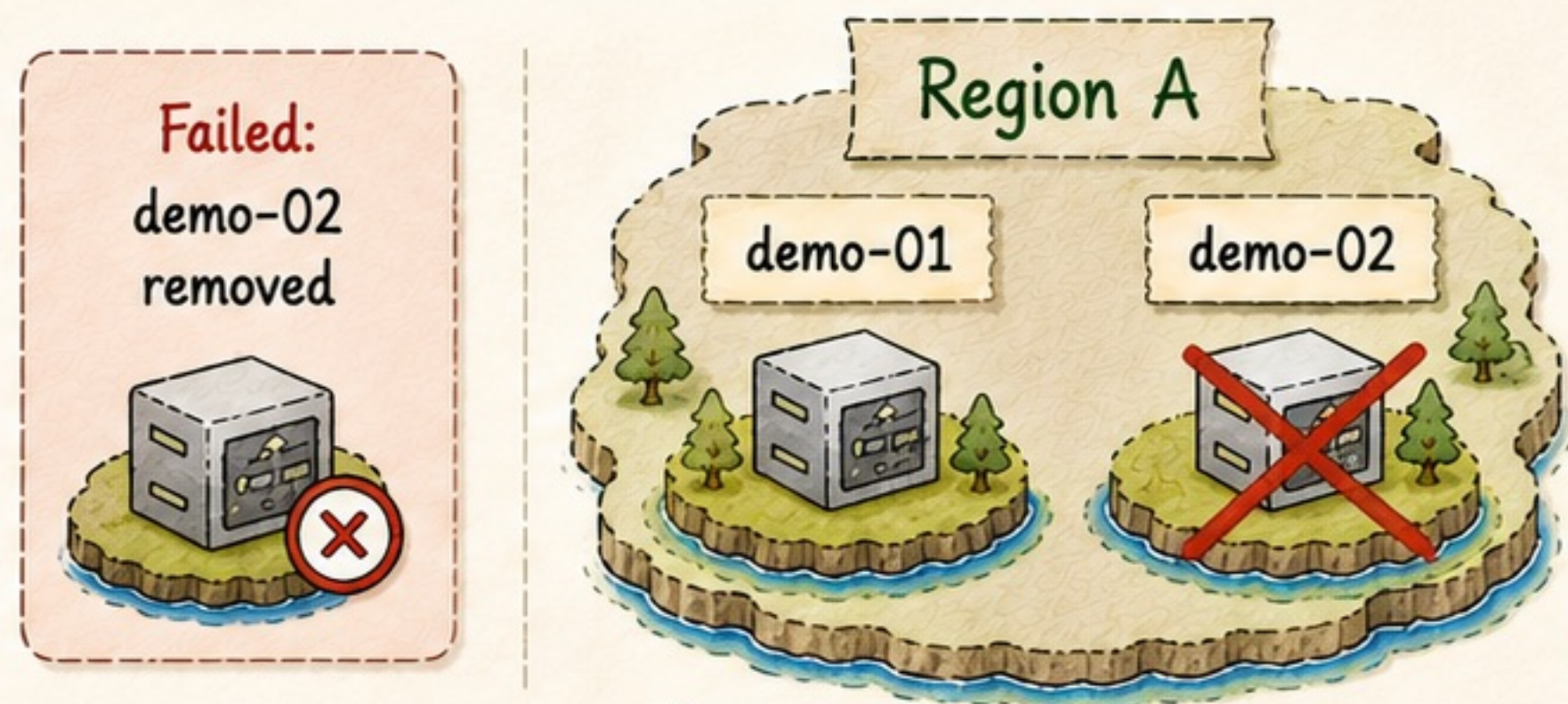


Cross-bucket behavior
is geo-layered DNS,
not one flat
Avi-style global
weight pool.

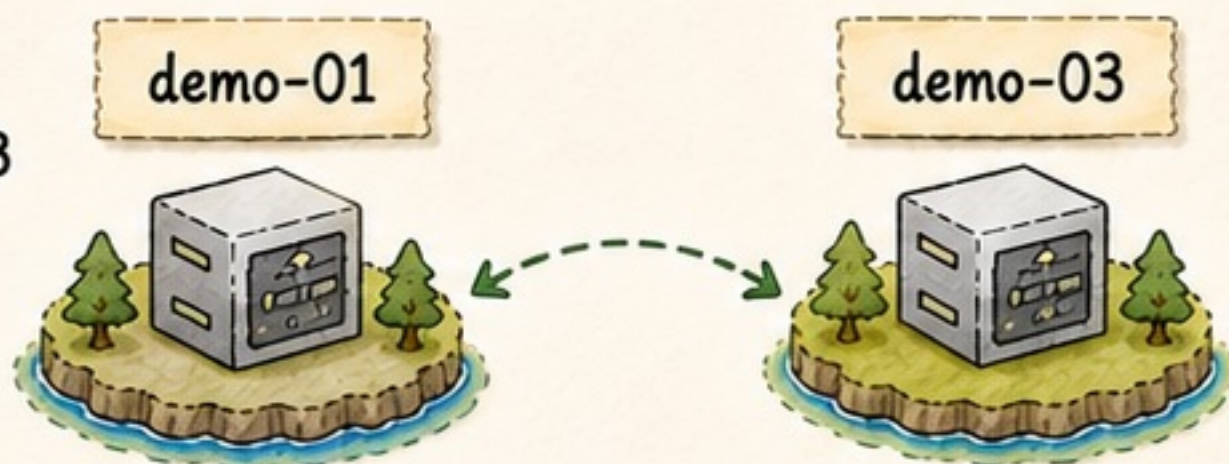
Scenario 2 Failure Behavior

Region member failures were survivable, but not Avi federation.

Region A member down



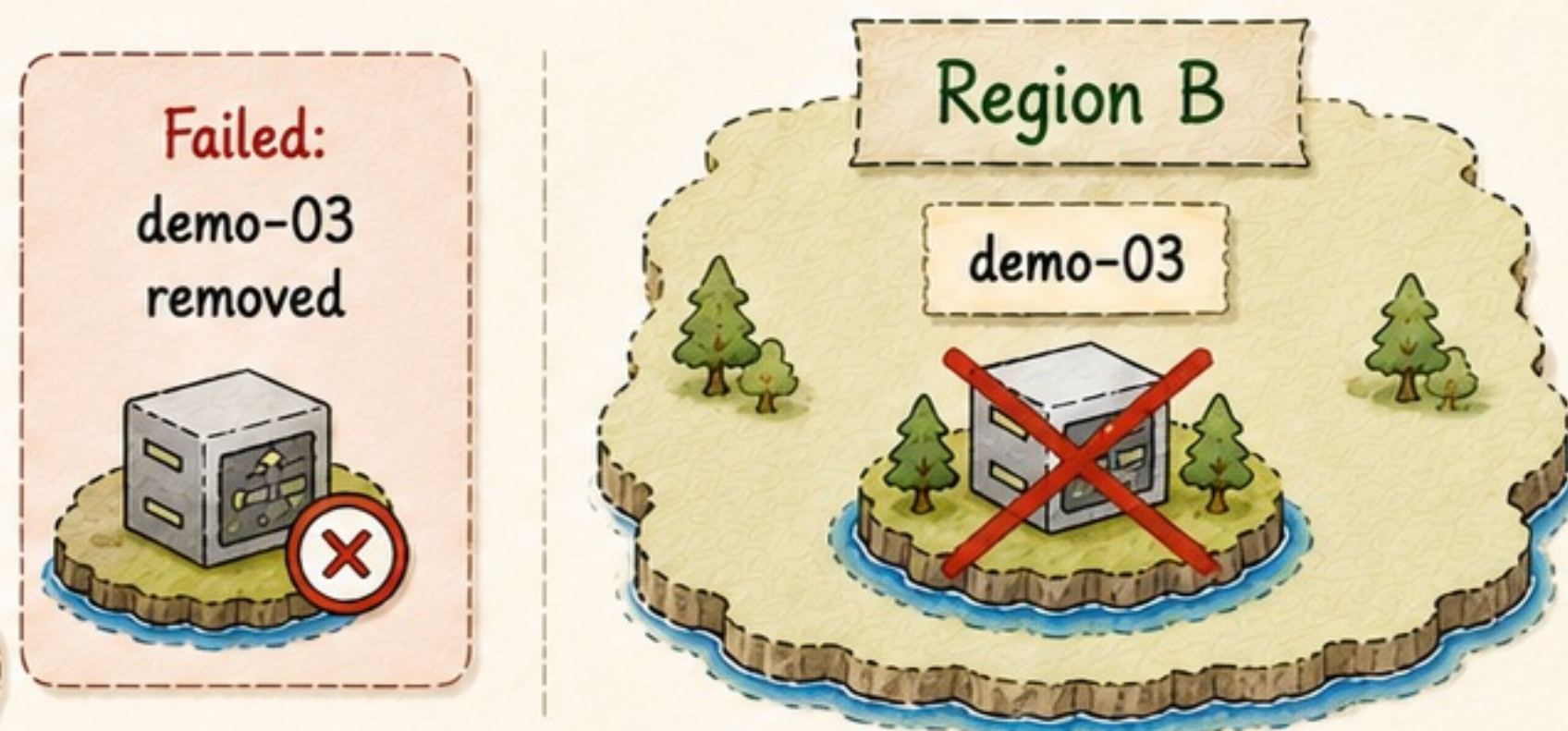
Remaining:
demo-01 + demo-03



Result: DNS answers survive

VS.

Region B member down



Remaining:
demo-01 + demo-02



Result: DNS answers survive



Validated survival; Avi-style site federation not implemented

Weight Is Not the Hard Failover Switch

DNSPolicy.weight expresses preference; active-groups expresses membership.

DNSPolicy.weight = 0



Soft
preference



Not reliable
enough alone
in this lab

VS.

active-groups remove



Hard
include/exclude
switch



Failed group
removed from
eligible DNS
answers



Production failover needs explicit pool membership control

Avi Replacement Assessment

RHCL replaces selected capabilities, not the entire Avi GSLB product.

Validated

- ✓ Multi-cluster DNS publication
- ✓ Same-bucket weighted distribution
- ✓ Health-based endpoint removal
- ✓ Per-hostname policy

Needs production design

- ! Failover SLA
- ! Controller reconciliation
- ! Multi-region semantics

Not proven in this PoC

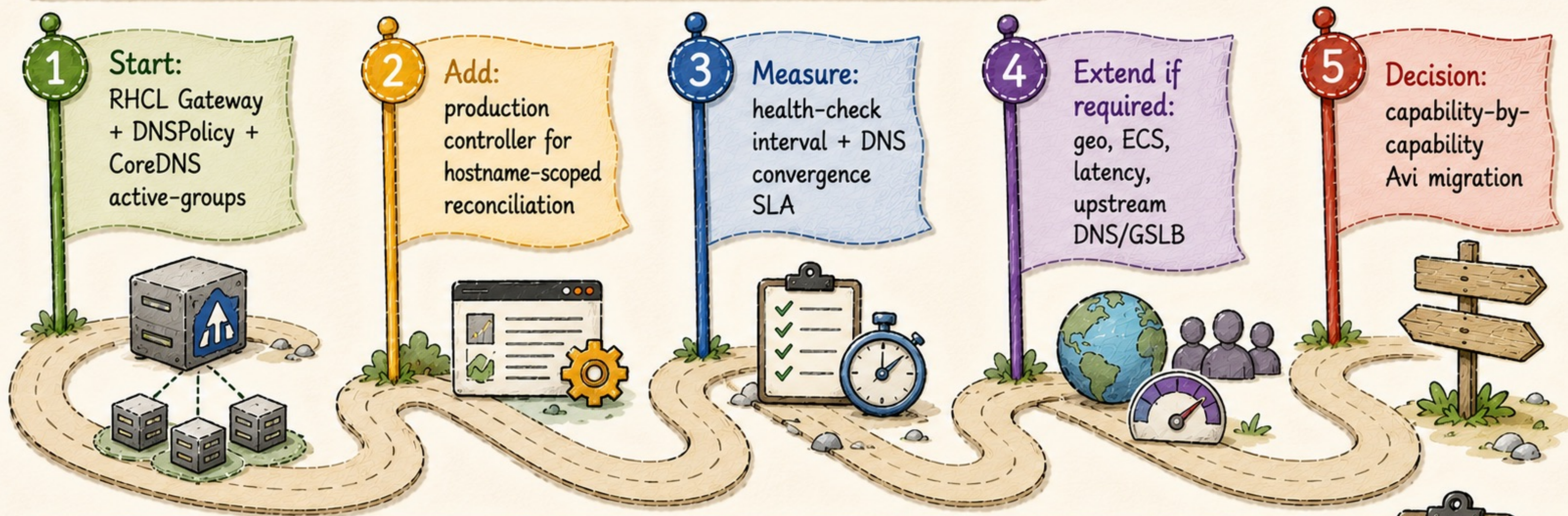
- ✗ Closest site routing
- ✗ Best-performing site routing
- ✗ EDNS Client Subnet routing
- ✗ Avi GSLB site federation



Capability-by-capability replacement, not a one-shot product swap.

Recommended Migration Path

Adopt RHCL where it is strong; design the missing GSLB layer explicitly.



Use RHCL for OpenShift-native DNS failover and weighted DNS; add extra design for full Avi-style GSLB.

